

1974 COMPUTER CENSUS

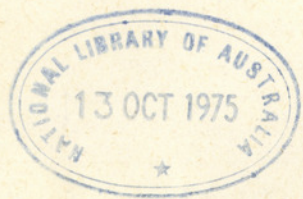
ELECTRONIC COMPUTERS IN AUSTRALIA
JUNE 1974

DEPARTMENT OF LABOR AND IMMIGRATION
MELBOURNE, 1975

1974

COMPUTER CENSUS

ANALYSIS OF THE 1974 CENSUS
RESULTS



RESEARCH REPORT NO. 1
1974

Electronic Computers in Australia — June 1974

The 13th annual census of computers in Australia was recently completed by the Department of Labor and Immigration. The census covers digital and hybrid computers (1). This article reports the situation existing at the end of June 1974. The statistical tables have been compiled from information provided by computer suppliers and, in some instances, computer users. The Department is very grateful for their co-operation.

It is difficult to determine whether some types of machines fall within the scope of the census, because of the continuous developments in computer hardware and in programmable electronic calculators and accounting machines. Single programme machines, sophisticated electronic calculators and electronic accounting machines have been excluded from the census. In general, visible record computers have also been excluded, although some larger types have been included. Externally programmed computers have also been included. The figures for both types of machines are shown separately in Table 3.

Previously published figures have been revised in some instances, as a result of the information which has been supplied for the current census.

Computers in Operation and on Order

There were 2420 digital computers in operation at 30 June 1974, a record increase of 429 over the previous 12 months. More than three-quarters of this increase (77.4 per cent) was accounted for by computers costing less than one hundred thousand dollars.

TABLE 1. Digital Computers Delivered, Removed and in Operation by Year*

	Calendar Years									
										Jan. — June
	1951—									
	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
Delivered during period	418	179	182	185	241	369	388	430	530	346
Removed during period	70	40	61	79	92	111	88	79	166	62
In operation at end of period	348	487	608	714	863	1,121	1,421	1,772	2,136	2,420

* Previously published figures for 1971 and 1972 have been revised.

The number of digital computers on order at the end of June 1974 reached the all-time high figure of 653, compared with 452 one year before. A very high proportion of these machines (61.7 per cent) were in the lowest cost-bracket, continuing the trend towards low-cost digital computers.

- (1) For the purposes of this census, hybrid computers are those with analogue and digital capabilities. They are shown separately in Tables 3,5,6,8 and 9.

TABLE 2. Digital Computers on Order at 30 June 1974 by Cost

	<i>Under \$100,000</i>	<i>\$100,000 to \$249,999</i>	<i>\$250,000 to \$499,999</i>	<i>\$500,000 to \$999,999</i>	<i>\$1,000,000 and over</i>	<i>Total</i>
No.	403	113	68	39	30	653
Percentage	61.7	17.3	10.4	6.0	4.6	

There were 2,400 digital stored programme computers in operation at census date. Externally programmed computers continue to be phased out and only 20 were known to be in operation at that date.

TABLE 3. Computers in Operation by Type*

Computer Type	<i>In Operation at End of Calendar Year</i>										<i>At 30 June</i>
	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	
Digital											
Stored programme	274	412	538	654	807	1,071	1,375	1,730	2,116	2,400	
Externally programmed	74	75	70	60	56	50	46	42	20	20	
TOTAL	348	487	608	714	863	1,121	1,421	1,772	2,136	2,420	
No. of larger visible record computers included in above totals	—	(3)	(19)	(29)	(43)	(52)	(66)	(65)	(59)	(57)	
Hybrid	3	4	5	9	22	41	53	63	67	82	

* Previously published figures for 1971 and 1972 have been revised.

Geographical Distribution

At 30 June 1974, New South Wales and Victoria together accounted for 72.3 per cent of the digital computers in operation and for 65.7 per cent of those on order.

TABLE 4. Digital Computers in Operation and on Order by Location*

<i>In operation at end of —</i>	<i>N.S.W.</i>	<i>Vic.</i>	<i>Qld</i>	<i>S.A.</i>	<i>W.A.</i>	<i>Tas.</i>	<i>N.T.</i>	<i>A.C.T.</i>	<i>Total</i>
1969	359	293	53	53	41	16	2	46	863
1970	471	358	79	73	54	19	3	64	1,121
1971	608	434	103	98	66	29	6	77	1,421
1972	750	554	131	117	82	33	7	98	1,772
1973	896	656	148	142	97	40	7	150	2,136
June 1974	1,008	741	163	172	106	48	7	175	2,420
%	(41.7)	(30.6)	(6.7)	(7.1)	(4.4)	(2.0)	(0.3)	(7.2)	
On order at 30 June 1974	234	195	70	43	40	13	—	58	653

* Previously published figures for 1971 and 1972 have been revised.

Of the 82 hybrid computers in operation at the end of June 1974, an increase of 11 over the past twelve months, 61 per cent were located in New South Wales or Victoria.

TABLE 5. Hybrid Computers in Operation and on Order by Location

	<i>N.S.W.</i>	<i>Vic.</i>	<i>Qld.</i>	<i>S.A.</i>	<i>W.A.</i>	<i>Tas.</i>	<i>N.T.</i>	<i>A.C.T.</i>	<i>Total</i>
In operation at 30 June 1974	30	20	8	12	3	3	—	6	82
On order at 30 June 1974	3	2	1	1	—	—	—	—	7

Cost Distribution

The proportion of digital computers costing less than one hundred thousand dollars increased further during the past twelve months, from 54.3 per cent to 58.4 per cent; over the last 5 years, it has increased by 19 percentage points.

The proportion of digital computers costing one million dollars or more fell slightly, from 5.7 per cent at end-June 1973 to 5.6 per cent at end-June 1974.

TABLE 6. Current Cost Distribution of Computers in Operation at 30 June 1974*

	<i>Under \$100,000</i>	<i>\$100,000 to \$249,999</i>	<i>\$250,000 to \$499,999</i>	<i>\$500,000 to \$999,999</i>	<i>\$1,000,000 and over</i>	<i>Total</i>
Digital No.	1,414	437	276	158	135	2,420
Digital %	(58.4)	(18.1)	(11.4)	(6.5)	(5.6)	
Hybrid No.	75	5	2	—	—	82
Hybrid %	(91.5)	(6.1)	(2.4)	—	—	

The current cost is the initial purchase cost plus the cost of any additions. An imputed cost has been used for leased equipment.

Industry Distribution

The industry distribution of the digital computers in operation and on order is presented in the tables which follow.

Table 7 shows that at census date the greatest number of digital computers in operation (789) were in the manufacturing sector, which accounted for 32 per cent of the total. Of these machines, 60.2 per cent were in the less than \$100,000 cost-bracket. The private sector as a whole accounted for 59.4 percent of the computers in operation.

Digital computers in operation costing one million dollars or more were evenly divided between the Government (68) and private (67) sectors; more were located in the Australian Government sector, the finance industry, data processing bureaux, metal manufacturing and transport and communication than in any other industry.

The proportion of computers on order in the private sector was 73.0 per cent of the total. The finance industry accounted for 12 out of the 30 machines costing one million dollars or more.

TABLE 7. Digital Computers in Operation by Year and Industry*

Industry	In Operation at End of Calendar Year										At 30 June
	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	
<i>Mining and Quarrying</i>	1	3	3	4	8	15	21	28	37	42	
<i>Manufacturing</i>											
Metal manufacturing, including machinery and electrical	42	63	88	101	119	146	182	256	315	358	
Clothing and textiles	11	15	20	20	35	37	40	46	44	50	
Food, drink and tobacco	11	12	21	27	30	42	63	76	86	93	
Pharmaceuticals, soap and toiletries	13	18	22	22	28	33	38	46	50	53	
Other chemicals, including oil	13	21	25	31	35	36	42	50	58	61	
Other manufacturing	21	29	43	54	64	78	103	124	149	174	
Total-Manufacturing	111	158	219	255	311	372	468	598	702	789	
<i>Electricity, Gas and Water</i>	8	14	17	22	26	32	41	45	50	54	
<i>Building and Construction</i>	8	10	12	11	12	15	18	25	26	28	
<i>Transport and Communication</i>	21	29	29	42	46	56	87	107	122	140	
<i>Finance</i>	43	46	53	69	82	106	126	156	176	186	
<i>Commerce</i>	29	36	44	49	50	63	79	83	101	108	
<i>Community and Business Services</i>											
Health	—	—	1	2	3	7	16	18	36	41	
Education	28	36	58	61	77	109	146	202	252	324	
Computer suppliers and data processing bureaux	40	54	63	68	87	122	140	167	211	235	
Other	7	16	17	19	21	29	36	40	43	52	
Total-Community and Business Services	75	106	139	150	188	267	338	427	542	652	
<i>Entertainment, Recreation, etc.</i>	1	4	3	5	7	11	17	17	20	24	
<i>Public Authority Activities</i> †											
Australian Government +	39	65	70	87	109	150	183	226	291	321	
State Government	2	2	2	3	4	7	11	25	27	33	
Local Government	10	14	17	17	20	27	32	35	42	43	
Total - Public Authority Activities	51	81	89	107	133	184	226	286	360	397	
TOTAL ALL INDUSTRIES	348	487	608	714	863	1,121	1,421	1,772	2,136	2,420	
Total Private	222	302	388	450	544	689	863	1,071	1,279	1,437	
Total Government x	126	185	220	264	319	432	558	701	857	983	

* Previously published figures for 1971 and 1972 have been revised.

† This segment refers to public authorities not elsewhere included.

+ Includes both administrative and research organisations and NASA installations in Australia.

x Includes, among others, universities and public hospitals.

TABLE 8. Digital and Hybrid Computers in Operation at 30 June 1974
by Industry and Current Cost*

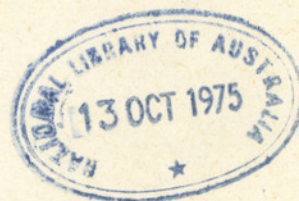
Industry	Digital Computers					Hybrid Total Computers	
	Under \$100,000	\$100,000 to \$249,999	\$250,000 to \$499,999	\$500,000 to \$999,999	\$1,000,000 and over		
Mining and Quarrying	28	6	3	2	3	42	—
Manufacturing							
Metal manufacturing, including machinery and electrical	221	63	35	25	14	358	2
Clothing and textiles	31	14	2	3	—	50	—
Food, drink and tobacco	55	21	15	2	—	93	—
Pharmaceuticals, soap and toiletries	25	18	8	2	—	53	—
Other chemicals, including oil	32	6	11	5	7	61	—
Other manufacturing	111	42	14	5	2	174	—
Total-Manufacturing	475	164	85	42	23	789	2
Electricity, Gas and Water	17	11	12	8	6	54	—
Building and Construction	7	14	4	3	—	28	—
Transport and Communication	65	27	25	9	14	140	2
Finance	48	38	41	34	25	186	—
Commerce	41	41	21	4	1	108	—
Community and Business Services							
Health	35	2	1	1	2	41	1
Education	268	26	11	9	10	324	69
Computer suppliers and data processing bureaux	115	50	32	17	21	235	3
Other	35	10	4	2	1	52	—
Total-Community and Business Services	453	88	48	29	34	652	73
Entertainment, Recreation, etc.	7	1	10	3	3	24	—
Public Authority Activities †							
Australian Government †	231	28	17	20	25	321	5
State Government	25	3	2	2	1	33	—
Local Government	17	16	8	2	—	43	—
Total-Public Authority Activities	273	47	27	24	26	397	5
TOTAL ALL INDUSTRIES	1,414	437	276	158	135	2,420	82
Total Private	774	317	184	95	67	1,437	4
Total Government x	640	120	92	63	68	983	78

* The current cost is the initial purchase cost plus the cost of any additions. An imputed cost has been used for leased equipment.

† This segment refers to public authorities not elsewhere included.

+ Includes both administrative and research organisations and NASA installations in Australia.

x Includes, among others, universities and public hospitals.



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TABLE 9. Digital and Hybrid Computers on Order at 30 June 1974
by Industry and Cost

Industry	Digital Computers					Total	Hybrid Computers
	Under \$100,000	\$100,000 to \$249,999	\$250,000 to \$499,999	\$500,000 to \$999,999	\$1,000,000 and over		
Mining and Quarrying	3	—	1	1	—	5	—
Manufacturing							
Metal manufacturing, including machinery and electrical	51	20	11	4	1	87	—
Clothing and textiles	13	6	2	—	—	21	—
Food, drink and tobacco	21	11	2	—	—	34	—
Pharmaceuticals, soap and toiletries	10	6	7	2	—	25	—
Other chemicals, including oil	18	2	2	1	1	24	—
Other manufacturing	25	4	5	1	1	36	—
Total-Manufacturing	138	49	29	8	3	227	—
Electricity, Gas and Water	5	8	1	2	1	17	—
Building and Construction	3	4	1	—	—	8	—
Transport and Communication	15	5	5	2	1	28	—
Finance	12	5	7	15	12	51	—
Commerce	31	10	12	3	—	56	—
Community and Business Services							
Health	1	3	—	—	—	4	—
Education	40	4	2	3	4	53	7
Computer suppliers and data processing bureaux	57	13	3	2	3	78	—
Other	32	6	5	—	1	44	—
Total-Community and Business Services	130	26	10	5	8	179	7
Entertainment, Recreation, etc.	4	—	—	—	3	7	—
Public Authority Activities †							
Australian Government +	54	3	—	3	1	61	—
State Government	1	—	—	—	1	2	—
Local Government	7	3	2	—	—	12	—
Total-Public Authority Activities	62	6	2	3	2	75	—
TOTAL ALL INDUSTRIES	403	113	68	39	30	653	7
Total Private	286	87	57	31	16	477	—
Total Government x	117	26	11	8	14	176	7

† This segment refers to public authorities not elsewhere included.

+ Includes both administrative and research organisations and NASA installations in Australia.

x Includes, among others, universities and public hospitals.



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